

Week 2 — Exploratory Data Analysis (EDA)

Junior Data Scientist Internship | Yuva Intern

Dataset: Online Retail

Analysis basis: Week 1 cleaned dataset

Focus: descriptive statistics, distributions, relationships, time patterns, product/country/customer analysis, outlier investigation, and hypothesis testing

Metric	Value
Rows	534,129
Columns after EDA feature creation	12
Duplicate rows	0
Negative-quantity transactions	9,251 (1.73%)
Regular sales revenue	10,642,110.80
Net recorded revenue	9,748,131.07
Identified customers	4,338
Overall AOV	479.56

Executive summary. The EDA treats unusual values as analytical information rather than automatically deleting them. Negative quantities were retained and analyzed separately; outliers were quantified using IQR and assessed through sensitivity analysis. Revenue-based analysis distinguishes regular positive-quantity sales from potentially reversal-related transactions.

1. EDA Methodology

The Week 2 analysis continues directly from the Week 1 cleaned Online Retail dataset. Revenue was derived as $\text{Quantity} \times \text{UnitPrice}$. Additional time features were created from InvoiceDate: YearMonth, DayOfWeek, and Hour. Regular-sales analyses use positive-quantity transactions, while negative-quantity transactions are examined separately.

Core analytical sequence: validation → descriptive statistics → revenue derivation → negative-quantity analysis → outlier analysis → distributions → correlation/scatter analysis → time trends → product/country/customer analysis → AOV → hypothesis testing → business insights.

2. Negative Quantity Analysis

Measure	Result
Negative transactions	9,251
Share of dataset	1.73%
Total negative quantity	-275,560.00
Revenue impact	-893,979.73
Regular sales revenue	10,642,110.80
Net recorded revenue	9,748,131.07

Interpretation: Negative-quantity transactions account for 1.73% of the cleaned dataset and reduce recorded revenue by 893,979.73. They were retained because they may represent returns, cancellations, credit transactions, or other reversals; the dataset does not provide enough information to classify every negative transaction with certainty. Removing them would risk overstating sales activity.

Analytical implication: regular-sales analysis uses positive quantities, whereas net revenue can include both positive and negative transaction effects.

3. Outlier Investigation

The IQR method was applied to Quantity, UnitPrice, and Revenue. Box plots and distribution plots were used to inspect the shape and spread of the variables. A temporary IQR-filtered version was also compared with the original data to evaluate the influence of extreme values.

Decision: outliers were not permanently removed. In retail data, large quantities, high prices, and high-value transactions can be legitimate business events. IQR filtering was therefore used as a sensitivity analysis rather than as an automatic cleaning rule.

4. Distribution and Relationship Analysis

Histograms and box plots were used for Quantity, UnitPrice, and Revenue. Scatter plots examined Quantity–UnitPrice, Quantity–Revenue, and UnitPrice–Revenue relationships. Correlation was interpreted as association rather than causation.

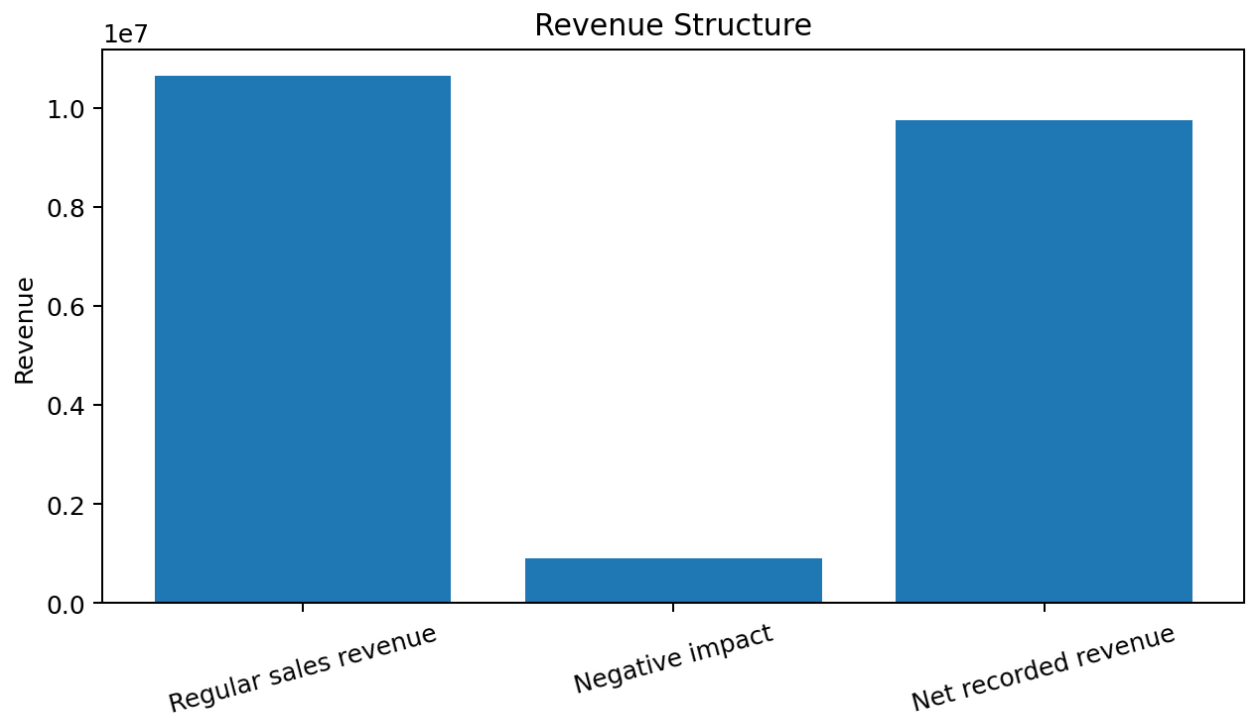
Important mathematical caveat: Revenue is constructed as $\text{Quantity} \times \text{UnitPrice}$. Consequently, correlations involving Revenue are partly induced by this formula and should not be interpreted as independent causal relationships.

5. Revenue Structure

The following comparison uses the exact metrics produced by the final notebook output.

Component	Amount
Regular sales revenue	10,642,110.80
Absolute negative-transaction impact	893,979.73
Net recorded revenue	9,748,131.07

The chart illustrates the relationship between regular-sales revenue, the magnitude of negative transaction impact, and net recorded revenue.



6. Time-Based Analysis

Monthly revenue trends and day-of-week patterns were examined using positive-quantity transactions. This separation prevents potential returns or reversals from being mixed directly into regular sales trends. The patterns describe the specific observation period and should not automatically be generalized to future periods.

7. Product and Country Analysis

The highest-revenue line item was **DOTCOM POSTAGE**, with recorded revenue of **206,248.77**. This item should be interpreted carefully: “DOTCOM POSTAGE” is a dataset line item/service charge rather than necessarily a physical merchandise product. Therefore, it should not be treated as a conventional product-demand finding without additional business classification.

United Kingdom had the highest recorded regular-sales revenue in the country-level analysis, at **9,001,744.09**. This describes the available dataset and does not establish profitability, market potential, or future performance.

8. Customer Analysis and AOV

Metric	Result
Customers with CustomerID	4,338
Identified-customer orders	18,532
Identified-customer revenue	8,887,208.89
Overall AOV	479.56

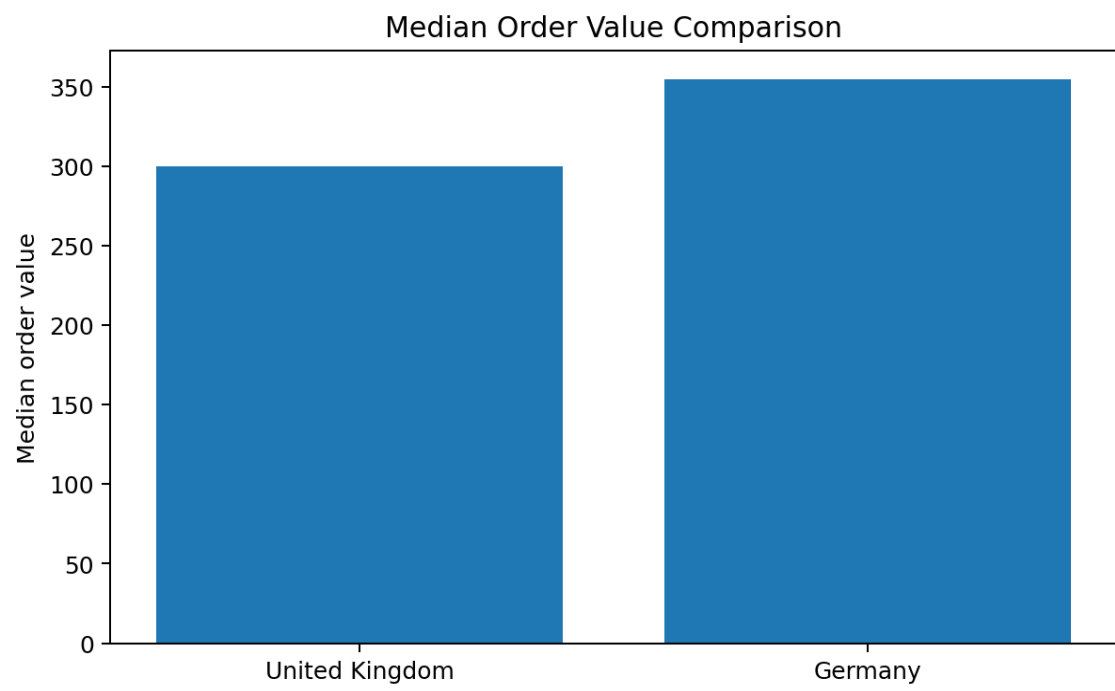
AOV was calculated at invoice level rather than transaction-line level. Customer analysis was restricted to records with a known CustomerID; missing CustomerIDs were not inferred.

9. Hypothesis Testing

The two countries with the highest regular-sale transaction counts were United Kingdom and Germany. Their order-value distributions were compared using a two-sided Mann–Whitney U test because retail order values can be skewed and contain extreme observations.

Statistic	Result
Group 1	United Kingdom
Group 2	Germany
United Kingdom orders	18,019
Germany orders	457
United Kingdom median	299.95
Germany median	354.85
U statistic	3,489,572.50
p-value	< 0.000001 (displayed as 0.000000)
Rank-biserial effect size	0.1525

At $\alpha = 0.05$, the p-value provides statistical evidence that the two observed order-value distributions differ. The rank-biserial effect size of 0.1525 is reported to describe magnitude alongside statistical significance. Statistical significance does not by itself establish commercial importance or causation.



10. Assumptions and Limitations

- Negative quantities may represent returns, cancellations, credits, or reversals; the dataset does not definitively classify every case.
- IQR outliers are statistical extremes, not automatically data errors.
- CustomerID values were not imputed because reliable customer identity could not be inferred.
- Revenue is not profit because costs, discounts, shipping expenses, and operating costs are unavailable.
- Correlation does not imply causation, and Revenue is mathematically derived from Quantity and UnitPrice.
- The hypothesis test describes the observed dataset; it does not establish that country causes differences in order value.
- The dataset covers a specific historical observation period, so findings should not automatically be generalized to other businesses or future periods.

11. Key Findings

- The cleaned EDA dataset contains 534,129 rows with zero duplicate rows.
- Negative-quantity transactions account for 1.73% of records and reduce recorded revenue by 893,979.73.
- Regular positive-quantity revenue is 10,642,110.80, while net recorded revenue after negative transaction effects is 9,748,131.07.
- Customer-level analysis identifies 4,338 customers and an invoice-level AOV of 479.56.
- The highest-revenue dataset line item is DOTCOM POSTAGE at 206,248.77; it should be interpreted as a line item/service charge rather than automatically as a physical product.
- United Kingdom has the highest recorded regular-sales revenue in this dataset at 9,001,744.09.
- The Mann–Whitney U test comparing United Kingdom and Germany order values produced $U = 3,489,572.50$, $p < 0.000001$, with rank-biserial effect size 0.1525.

12. Conclusion

The Week 2 EDA demonstrates a structured approach to understanding retail transaction data while explicitly connecting data-handling choices to analytical consequences. The analysis preserves potentially meaningful negative transactions, investigates rather than automatically deletes outliers, separates regular sales from reversal-like activity, and combines descriptive statistics with visualization and non-parametric hypothesis testing. These decisions improve interpretability while making the assumptions and limitations of the analysis explicit.